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AdVantage OS: An AI-Driven Local Growth Platform for SMEs

FINAL YEAR PROJECT WORK (BIS786)

SYNOPSIS

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1.Title of the Project

AdVantage OS: An AI-Driven Local Growth Platform for SMEs

2. Introduction

Small and medium enterprises often struggle to respond quickly to fast-changing local customer preferences because they usually lack access to integrated analytics, campaign intelligence, and operational forecasting tools that larger firms can afford. AdVantage OS is proposed as a unified Growth-as-a-Service platform that translates local social media activity into practical marketing and inventory decisions for SMEs, effectively connecting online trend discovery with offline retail execution. This system is particularly relevant in a hyper-local digital economy where neighborhood-level interests can strongly influence footfall, product demand, and promotional performance. By combining trend detection, automated campaign generation, influencer discovery, and inventory monitoring within a single platform, the project addresses both growth and operational resilience needs of local retailers. The initiative is important because most existing SME marketing tools focus only on campaign creation or only on analytics; AdVantage OS closes the loop from social listening to action, performance tracking, and stock readiness, making it valuable for both academic study and real-world application in sectors such as cafés, boutiques, and salons.

3. Literature Survey

A review of the available references and project materials shows that the proposed work stands at the intersection of retail forecasting, social media trend analysis, recommender systems, and multimodal advertising intelligence. The presentation cites recent studies on demand forecasting, predictive sales analysis, social media trend prediction, topic modeling, and AI-assisted advertising, which together provide the conceptual base for the project architecture.

➤ **Paper 1 : Agentic AI-Driven Demand Forecasting in Retail Systems**

- **Results :** Agentic AI and predictive analytics significantly improve retail demand forecasting accuracy, reducing forecast error, stockouts, and enabling better inventory and pricing decisions for large retail systems.
- **Limitations :** The work is focused on back-end forecasting for larger or enterprise retailers and does not integrate social media trend discovery, influencer analytics, or creative campaign workflows needed to close the loop for SMEs.

➤ **Paper 2 : AI in Social Media Marketing Using Machine Learning to Analyze Trends and Predict Consumer Sentiment**

- **Results :** ML-based analysis of social media features can predict consumer sentiment and social trends, helping brands understand audience reactions and optimise social media content strategy.

- **Limitations :** It remains limited to marketing insight and content optimisation, without linking trend and sentiment outputs to inventory management or providing an end-to-end workflow from social listening to store-level operational decisions.
- **Paper 3 : AI Integrated Recommender System for Improving E-commerce Experience**
 - **Results :** AI-based recommender systems personalise product suggestions based on user behaviour and preferences, improving customer experience, engagement, and sales in e-commerce platforms.
 - **Limitations :** The solution is confined to online recommendation quality and personalisation, ignoring hyper-local social trends, micro-influencer selection, and physical store inventory constraints that matter for SME retail.
- **Paper 4 : InvadeAI: Interactive Advertising with Multimodal AI**
 - **Results :** InvadeAI uses multimodal AI to generate rich, interactive advertisements that adapt to user context, making creatives more dynamic and improving customer engagement across digital channels.
 - **Limitations :** The system focuses on creative generation and interactive ad experiences only, without coupling ads to local demand forecasts, social trend lifecycles, or stock and inventory conditions for SMEs, leaving the execution loop incomplete.
- **Paper 5 : Smart Retail: Utilizing Machine Learning for Demand Prediction, Price Strategy, and Inventory Management**
 - **Results :** ML-based methods for demand prediction, dynamic pricing, and inventory optimisation improve profit margins, reduce stockouts, and make replenishment more efficient in smart retail operations.
 - **Limitations :** The framework is largely enterprise-oriented, assuming rich structured data and infrastructure, and does not integrate live social media trends, influencer activity, and creative campaign tools into a unified, SME-friendly end-to-end platform.

4. Problem Statement

Small and medium enterprises often struggle to respond quickly to fast-changing local customer preferences because they usually lack access to integrated analytics, campaign intelligence, and operational forecasting tools that larger firms can afford. AdVantage OS is proposed as a unified Growth-as-a-Service platform that translates local social media activity into practical marketing and inventory decisions for SMEs, effectively connecting online trend discovery with offline retail execution. This system is especially relevant in a hyper-local digital economy where neighborhood-level interests can strongly influence footfall, product demand, and promotional performance. By combining trend detection, automated campaign generation, influencer discovery, and inventory monitoring within a

single platform, the project addresses both growth and operational resilience needs of local retailers. Unlike most existing SME marketing tools that focus only on campaign creation or analytics, AdVantage OS closes the loop from social listening to action, performance tracking, and stock readiness, making it valuable for academic study and real-world application in sectors such as cafés, boutiques, and salons.

5. Objective of the Proposed Project

The major objectives of the proposed project are:

- Detect and quantify hyper-local trends: monitor social platforms (Reddit, YouTube, etc.) to identify locality-specific topics, convert them into structured neighbourhood- or PIN-level trends, and forecast each trend's growth, peak, and decay with time-series models. Present easy-to-read scores (trend strength, local relevance, urgency) so non-technical owners can quickly prioritise opportunities.
- Drive targeted campaigns and influencer selection: enable SMEs to generate on-brand images and captions with generative AI, estimate on-device CTR/engagement before publishing, and improve ROI by clustering influencers by engagement, audience size, niche, and locality while filtering suspicious/botted accounts.
- Close the loop with inventory-aware execution: capture receipts via OCR and use ML on sales velocity and stock levels to predict stockout risk for trending products; send proactive WhatsApp alerts for “trend high, stock low” conditions and provide a unified, modular dashboard that integrates trend intelligence, influencer discovery, creative generation, and inventory analytics.

6. Scope of the Project

The project targets SMEs such as cafés, boutiques, and salons, beginning with a pilot in selected neighbourhoods of Bengaluru. It ingests public social trend data from sources like Reddit and YouTube (with Instagram planned for future extension) and feeds them into a Trend Engine that applies text preprocessing, LDA topic modelling, and basic time-series forecasting to identify and score locality-specific trends. An Influencer Discovery module clusters creators by engagement rate, audience size, niche, and locality while applying simple rules to filter suspicious or bottled accounts. A Creative Studio web dashboard enables SMEs to generate trend-aligned ad images and captions using generative models and provides a basic CTR/engagement estimate before campaigns are launched. The Inventory Edge module uses OCR on receipt images to update stock levels and, combined with sales and trend signals, triggers proactive WhatsApp alerts when products risk stocking out, thereby closing the loop between campaigns and inventory. The system is delivered as a prototype-level modular full-stack application (frontend, backend, database, ML components) that follows basic security best practices; advanced production features—such as multi-tenant billing, large-scale scalability, dynamic pricing, and detailed sustainability or supplier modules—are intentionally kept out of scope for this project.

7. Methodology

Follow a modular, phased development approach: first build and validate each core module (Trend Engine, Influencer Discovery, Creative Studio, Inventory Edge), then integrate them into a single dashboard.

Detailed Flow Chart

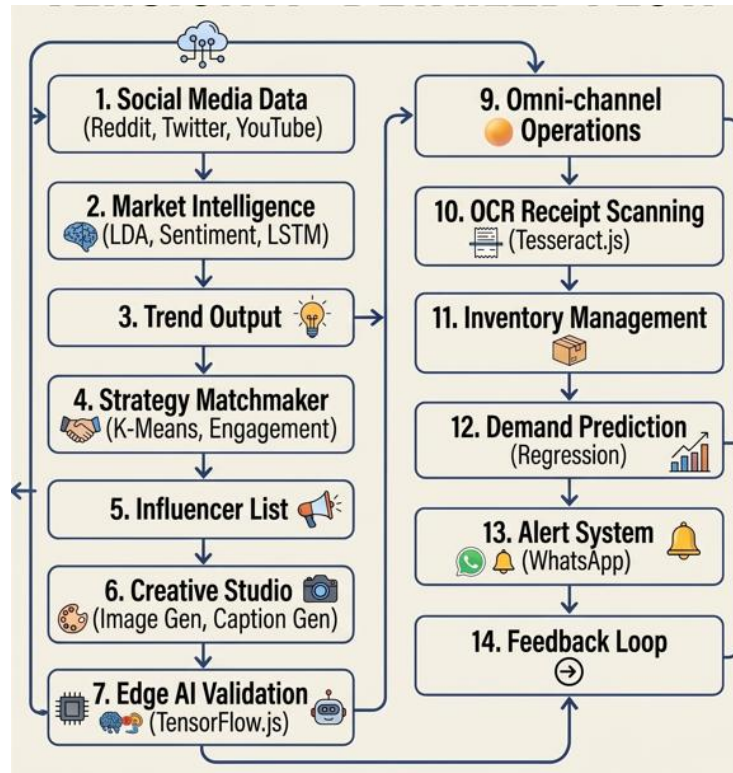


Figure 1.1 : System Workflow Diagram

1. Data collection and storage

- Identify relevant subreddits, YouTube channels, and keyword filters that represent local interests for the chosen SME niche (e.g., cafés, salons, boutiques in Bengaluru).
- Use official APIs (Reddit, YouTube) to periodically fetch posts and video metadata (titles, descriptions, comments, timestamps, basic engagement).
- Clean and normalise the raw data (remove duplicates, strip HTML/emojis, handle language issues) and store it in a PostgreSQL/Supabase database with clear schemas for trends, posts, influencers, campaigns, and inventory records.

2. Trend detection and forecasting (Trend Engine)

- Perform text preprocessing: tokenisation, lowercasing, stop-word removal, lemmatisation/stemming on social media text.
- Apply LDA topic modelling to cluster posts into coherent topics/niches representing potential trends.
- For each topic, build a time series of activity (e.g., posts per day, engagement per day) and train an LSTM or similar forecasting model to estimate future strength of that trend over a short horizon.
- Compute a “window of opportunity” score using predicted peak time, current growth rate, and locality relevance, and expose this as a simple numeric or colour-coded indicator in the dashboard.

3. Influencer discovery and validation (Matchmaker)

- Collect candidate influencer profiles (e.g., from YouTube, Instagram or sample dataset) along with features such as follower count, engagement rate, content tags, and location.
- Standardise and scale these features, then use K-Means clustering to group influencers by niche, size, and engagement pattern.
- Implement bot/fraud detection rules (e.g., abnormal follower/following ratio, very high likes with very few genuine comments, repetitive content) and optionally a small classifier to flag risky accounts.
- For each active trend, compute an influencer–trend similarity score (based on niche tags and audience locality) and recommend a shortlist of micro-influencers to the SME user.

4. Creative generation and CTR estimation (Creative Studio)

- Build a React + Tailwind CSS dashboard where users can pick a trend and product, then request creative suggestions.
- Integrate a text-to-image model (e.g., Stable Diffusion or similar API) to generate ad images from prompts derived from the selected trend and brand style.
- Use a language model to generate multiple caption options aligned with the trend’s tone and target audience.
- Run a TensorFlow.js model in the browser to classify the product/creative into theme categories and estimate expected CTR or engagement, allowing users to compare and choose between creatives.

5. Receipt OCR and inventory tracking (Inventory Edge)

- Enable SME users to upload or capture receipt images through the web interface or a simple mobile-friendly view.
- Use OCR (e.g., Tesseract.js) to extract key fields from receipts: item name, quantity, price, date/time.
- Map extracted items to internal product IDs, update stock-in/stock-out records, and maintain approximate real-time inventory levels for monitored SKUs.
- Handle basic data cleaning for OCR errors (e.g., fuzzy matching of item names, manual correction option in the UI).

6. Predictive stockout alerts and messaging

- Combine trend forecasts, campaign status (active creatives/influencers), and recent sales velocity to compute future demand estimates per product.
- Train a simple regression or classification model that predicts whether a product is at risk of stockout within a given time window if current conditions continue.
- When risk crosses a threshold, trigger an automated WhatsApp alert to the business owner using a messaging API, including a clear message like “Product X may stock out in ~3 days during Trend Y – consider reordering.”
- Log alerts and business responses (e.g., “ignored”, “reordered”) for future tuning of the alert thresholds and models.

7. System integration and architecture

- Design a RESTful backend (e.g., Python + FastAPI/Flask) exposing endpoints for trend data, influencer recommendations, creative generation requests, OCR uploads, inventory metrics, and alerts.
- Implement authentication and basic access control (e.g., using Supabase Auth or JWT) so each SME sees only its own campaigns and inventory data.
- Connect the React frontend to backend APIs using a modular service layer, ensuring each module (Trend Engine, Matchmaker, Factory, Storefront) appears as a coherent section in the dashboard.
- Use background jobs or schedulers (cron/queues) for periodic data ingestion, model retraining, and batch computations (trend updates, influencer scoring, inventory projections).

Modular Flow Chart

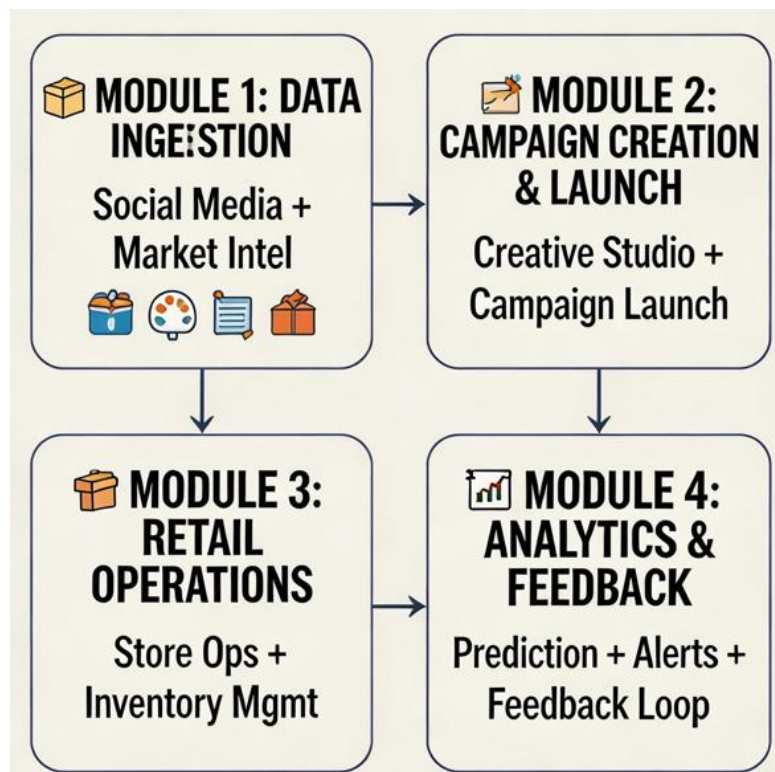


Figure 1.2 : Modular System Flowchart

- Module 1 – Data Ingestion collects social media and market intelligence signals, forming the raw input for trend discovery and downstream analytics.
- Module 2 – Campaign Creation & Launch uses a Creative Studio to design ads and then launches campaigns aligned to the detected trends.
- Module 3 – Retail Operations connects the system to store operations and inventory management so that promotions are grounded in actual stock levels.
- Module 4 – Analytics & Feedback analyses campaign and sales data, generates predictions and alerts, and feeds insights back to refine future campaigns and operations.

8. Evaluation and testing

- Evaluate topic quality using qualitative inspection and optional coherence metrics, and verify that detected topics correspond to recognisable trends.
- Measure forecast accuracy using standard time-series metrics (e.g., MAE, RMSE) on held-out trend data.
- Assess **influencer recommendation quality** based on engagement metrics and manual judgement of niche/locality match.

- Validate OCR accuracy and inventory predictions using sample receipts and manually tracked stock levels, and measure how early alerts are generated before actual stockouts.
- Conduct user testing with a small number of SME-like users (or classmates role-playing shop owners) to check whether the dashboard flow is understandable and useful.

Requirements

Frontend

- React.js for building the dashboard UI and campaign workflows.
- Tailwind CSS for responsive, component-based styling of charts, cards, and forms.
- JavaScript/TypeScript (ES6+) for interactive components, API calls, and state management on the client side.

Backend

- Python + FastAPI/Flask for implementing REST APIs, ML orchestration, and business logic.
- Supabase (PostgreSQL + Auth) for persistent storage of users, trends, influencers, campaigns, and inventory, with built-in authentication.
- REST APIs for communication between frontend, ML services, and external providers (AI models, WhatsApp).

AI & ML Tools

- scikit-learn for LDA topic modeling, K-Means clustering, and regression models (CTR and inventory prediction).
- TensorFlow / Keras for LSTM or alternative time-series forecasting models for trend strength over time.
- Transformers (Hugging Face) for sentiment analysis and text embeddings used in trend scoring and caption generation.
- TensorFlow.js for on-device image classification and CTR estimation directly in the browser.

Generative AI Tools

- Stable Diffusion / DALL-E via Hugging Face or OpenAI APIs for generating ad images based on trend-aware prompts.
- GPT-style language models for generating campaign captions, influencer outreach messages, and copy variants.

OCR & Document Tools

- Tesseract.js for in-browser OCR on receipt images to extract items, quantities, and totals.
- Basic image preprocessing utilities (e.g., resizing, grayscale, thresholding) to improve OCR accuracy.

Messaging & Integration

- Twilio WhatsApp Business API for sending automated stock-risk alerts and campaign notifications to SME owners.
- Cron jobs / background workers (e.g., via server schedulers or Supabase functions) for periodic data ingestion and model updates.

Datasets

- Social media datasets: Public Reddit posts, YouTube video data (titles, descriptions, comments, engagement), Google Search results (trending queries, related topics), and Google Trends time-series signals, all fetched via official APIs or export tools, used for local trend detection and forecasting.
- Influencer dataset: Profile and engagement data for micro-influencers collected via the YouTube Data API and Instagram creator profiles scraped through Apify (followers, likes, comments, niche tags, location), curated into structured CSV/JSON formats.
- M5 Forecasting – Walmart sales data: Public Walmart retail sales time series dataset (M5 Forecasting competition) used to experiment with and benchmark LSTM based demand forecasting models before adapting them to local SME trend data.
- A Million News Headlines: Large scale news headline corpus used as an additional text source to prototype and validate LDA topic modelling pipelines prior to training on domain specific Reddit/YouTube text.

8. Expected Outcome

The expected outcome of the project is a functional prototype of an AI-driven local growth operating system that demonstrates how social trend intelligence, campaign support, influencer matching, and retail operations can be integrated into a single SME-oriented platform. The system is expected to help local businesses identify meaningful opportunities earlier, prepare better campaigns faster, and improve readiness for trend-driven demand fluctuations.

From a technical standpoint, the project will produce a modular full-stack application, trained or integrated ML components, a dashboard interface, and a workflow that links data ingestion, prediction, creative generation, and inventory alerts. The work is also expected to contribute academically by showing a practical multimodal AI architecture that combines NLP, forecasting, computer vision, OCR, and generative AI in one coherent retail use case.

The likely deliverables include a web application, trained or configured machine learning modules, API integrations, demonstration datasets, experiment results, and project documentation. The project may further open pathways for later enhancements such as dynamic pricing, sustainability analytics, geo-local heatmaps, and cross-channel campaign automation.

9. Work Plan Timeline

Phase	Task	Duration
Phase 1	Problem understanding, literature review, and requirement analysis	Week 1-2
Phase 2	System design, dataset planning, and architecture finalization	Week 3-4
Phase 3	Social data ingestion module and database setup	Week 5-6
Phase 4	Trend analysis, forecasting, and influencer discovery modules	Week 7-9
Phase 5	Creative Studio and dashboard development	Week 10-11
Phase 6	OCR-based receipt processing, inventory prediction, and alert module	Week 12-13
Phase 7	Integration, testing, evaluation, and performance refinement	Week 14-15
Phase 8	Final documentation, presentation preparation, and project review	Week 16

This timeline follows the template expectation of a phased academic project plan while aligning the sequence with the proposed closed-loop system design. It also leaves room for iterative testing and module-wise integration during the semester.

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